ABSTRACT OF THE DISCLOSURE

A communication protocol for negotiating data encoding schemes at the Physical (Layer 1) and Data-Link (Layer 2) layers of the OSI model, specifically for high-cardinality parallel interfaces. The protocol comprises: a device, upon connection, transmitting a capabilities frame using a universally supported, low-cardinality default encoding; the capabilities frame advertising the device's supported encoding maps (e.g., "UTF-8-Retrofit," "Machine-Code-24bit"); a receiving device responding with a selection frame choosing one of the advertised encodings; and both devices then switching from the default encoding to the selected, high-density encoding for all subsequent data transmission in the session. This moves encoding negotiation from the Application Layer to the hardware layer, a necessary innovation for interfaces where the physical signal representation itself is variable.